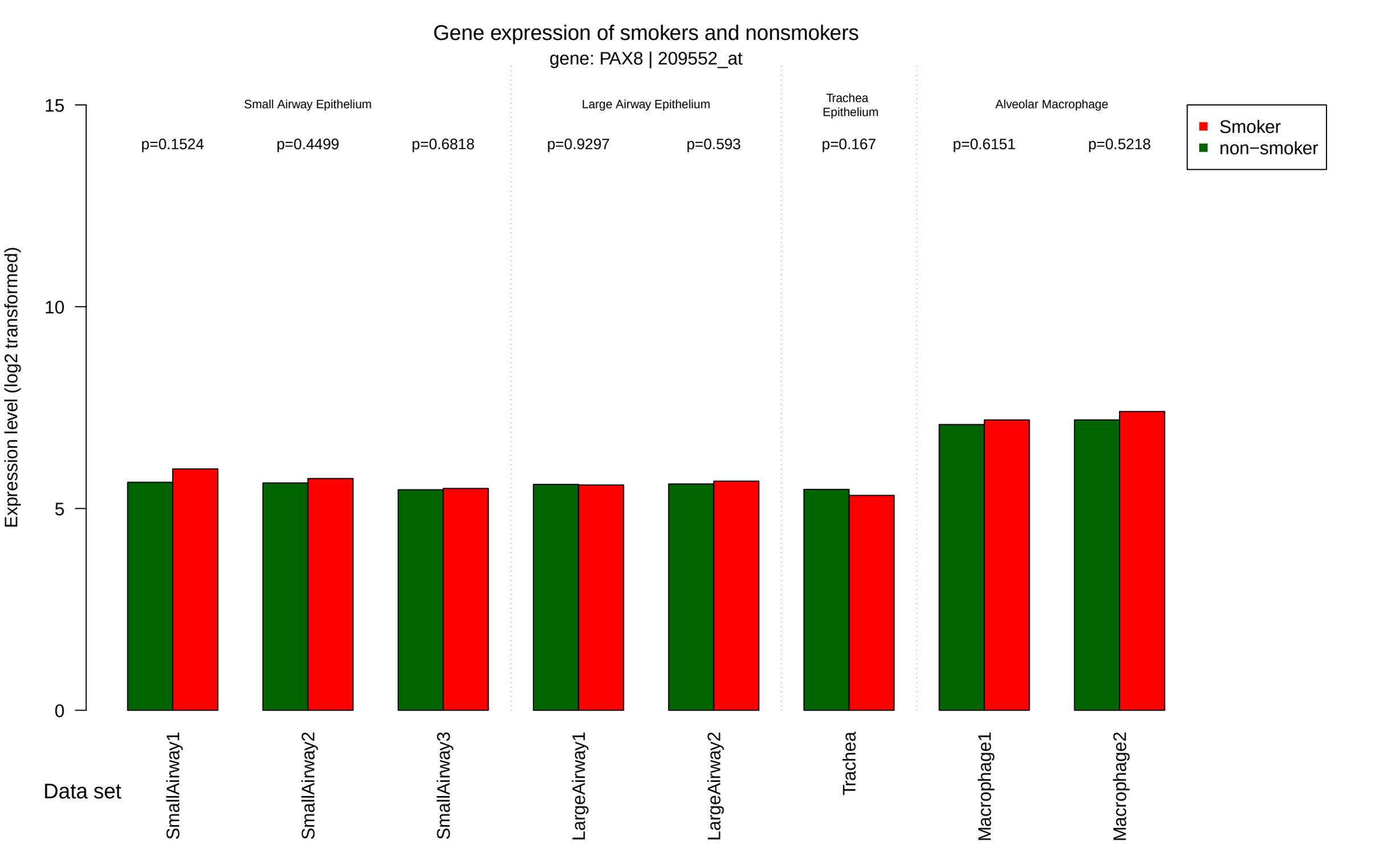
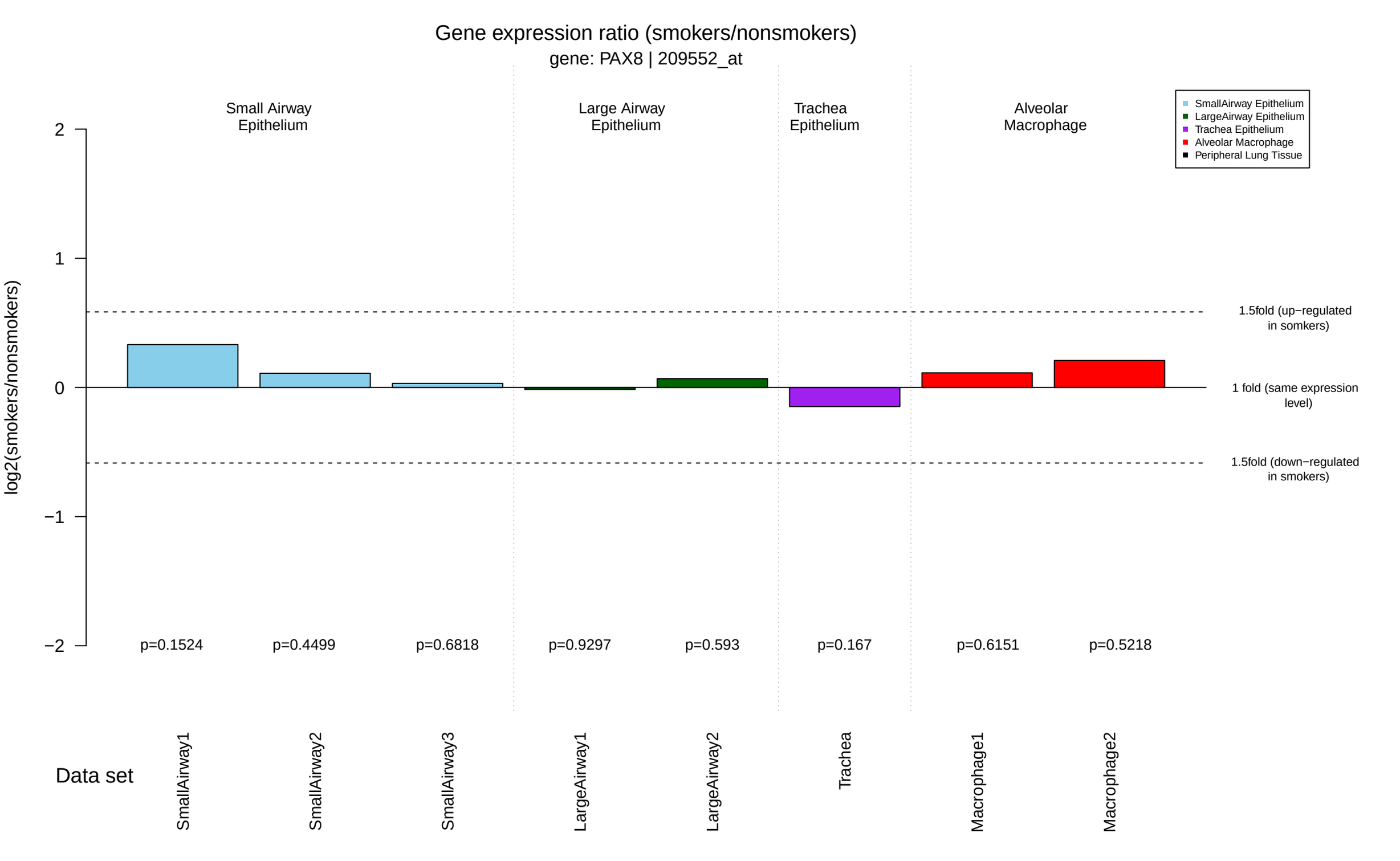
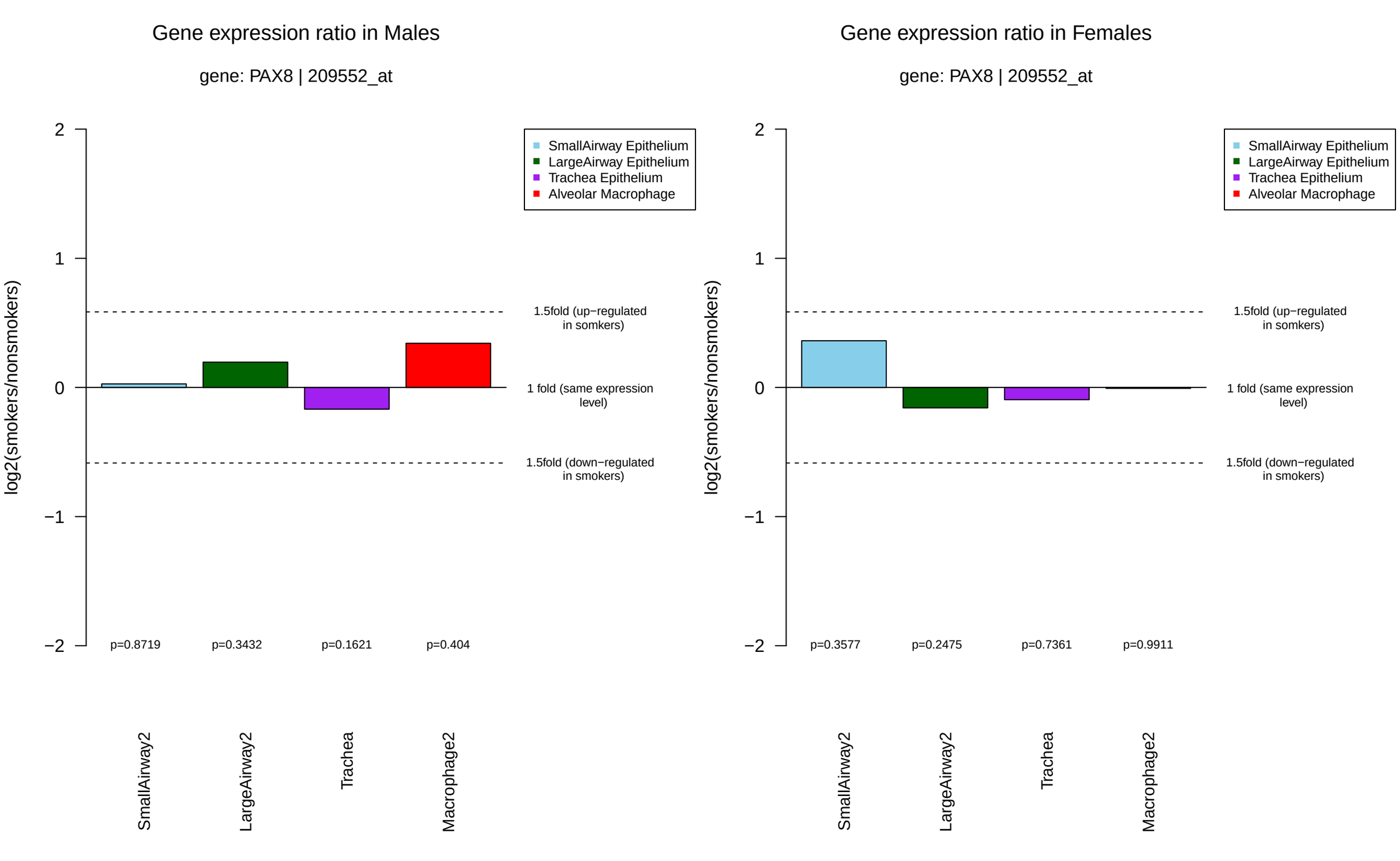
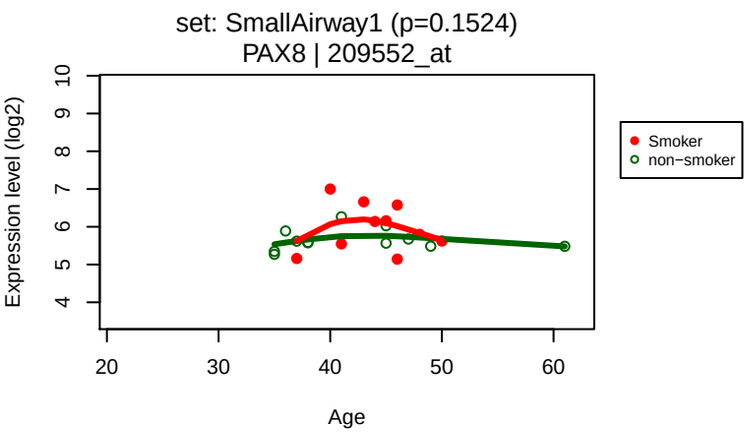
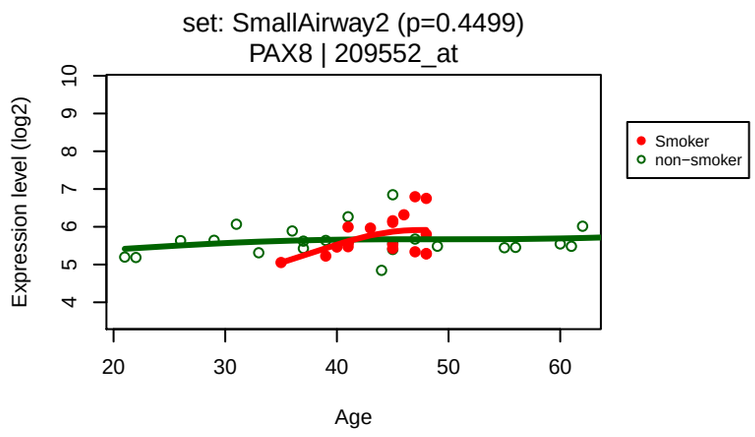
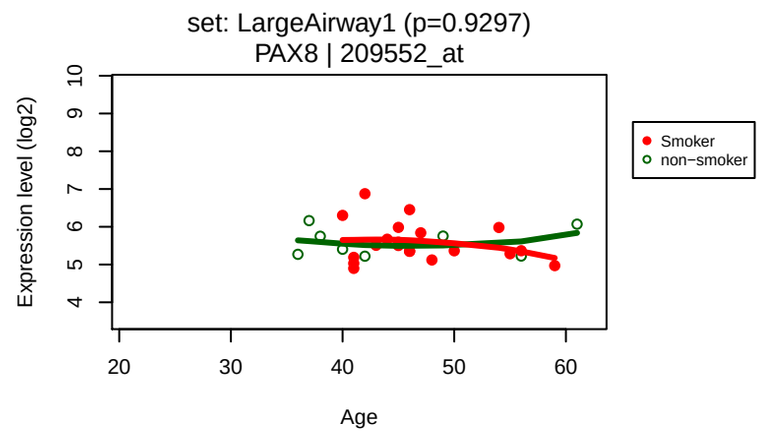

Gene expression of smokers and nonsmokers
gene: PAX8 | 209552_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1524
p=0.4499
p=0.6818
p=0.9297
p=0.593
p=0.167
p=0.6151
p=0.5218
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

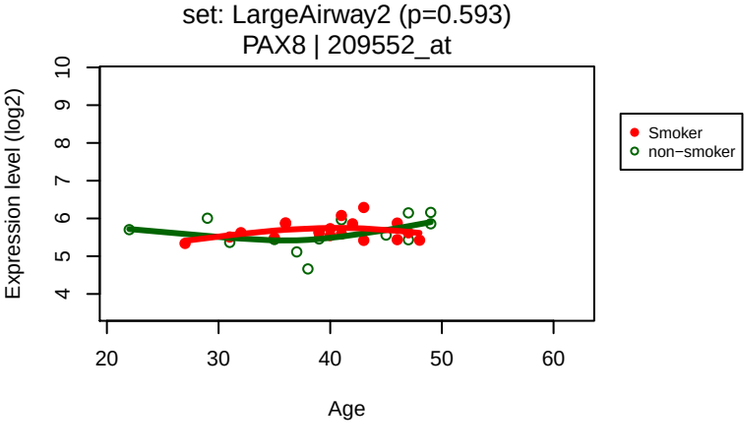

Gene expression ratio (smokers/nonsmokers)
gene: PAX8 | 209552_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1524
p=0.4499
p=0.6818
p=0.9297
p=0.593
p=0.167
p=0.6151
p=0.5218
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

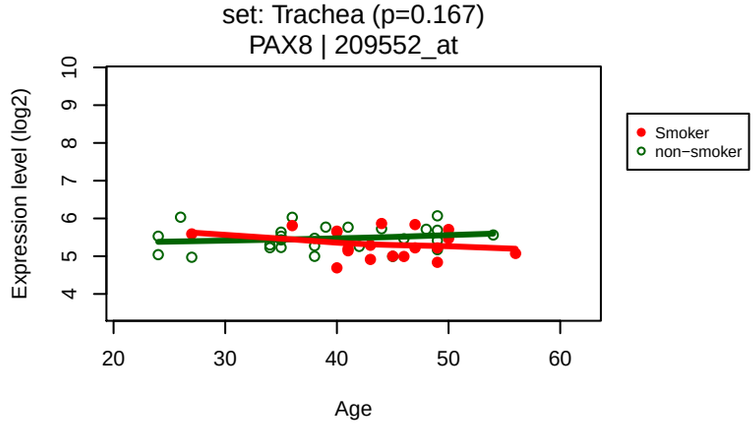

Gene expression ratio in Males
gene: PAX8 | 209552_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8719
p=0.3432
p=0.1621
p=0.404
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAX8 | 209552_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3577
p=0.2475
p=0.7361
p=0.9911
SmallAirway2
LargeAirway2
Trachea
Macrophage2

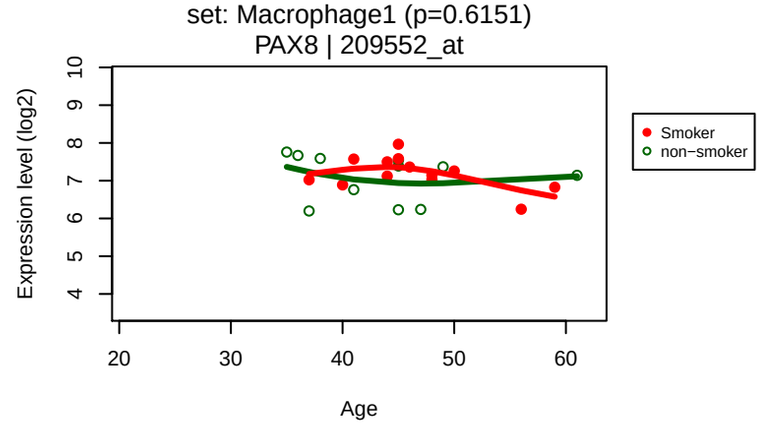

set: SmallAirway1 (p=0.1524)
PAX8 | 209552_at
Expression level (log2)
Age
Smoker
non−smoker

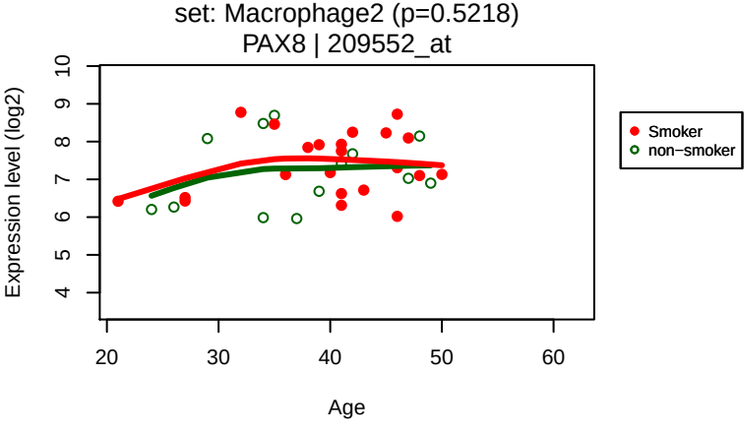

set: SmallAirway2 (p=0.4499)
PAX8 | 209552_at
Expression level (log2)
Age
Smoker
non−smoker

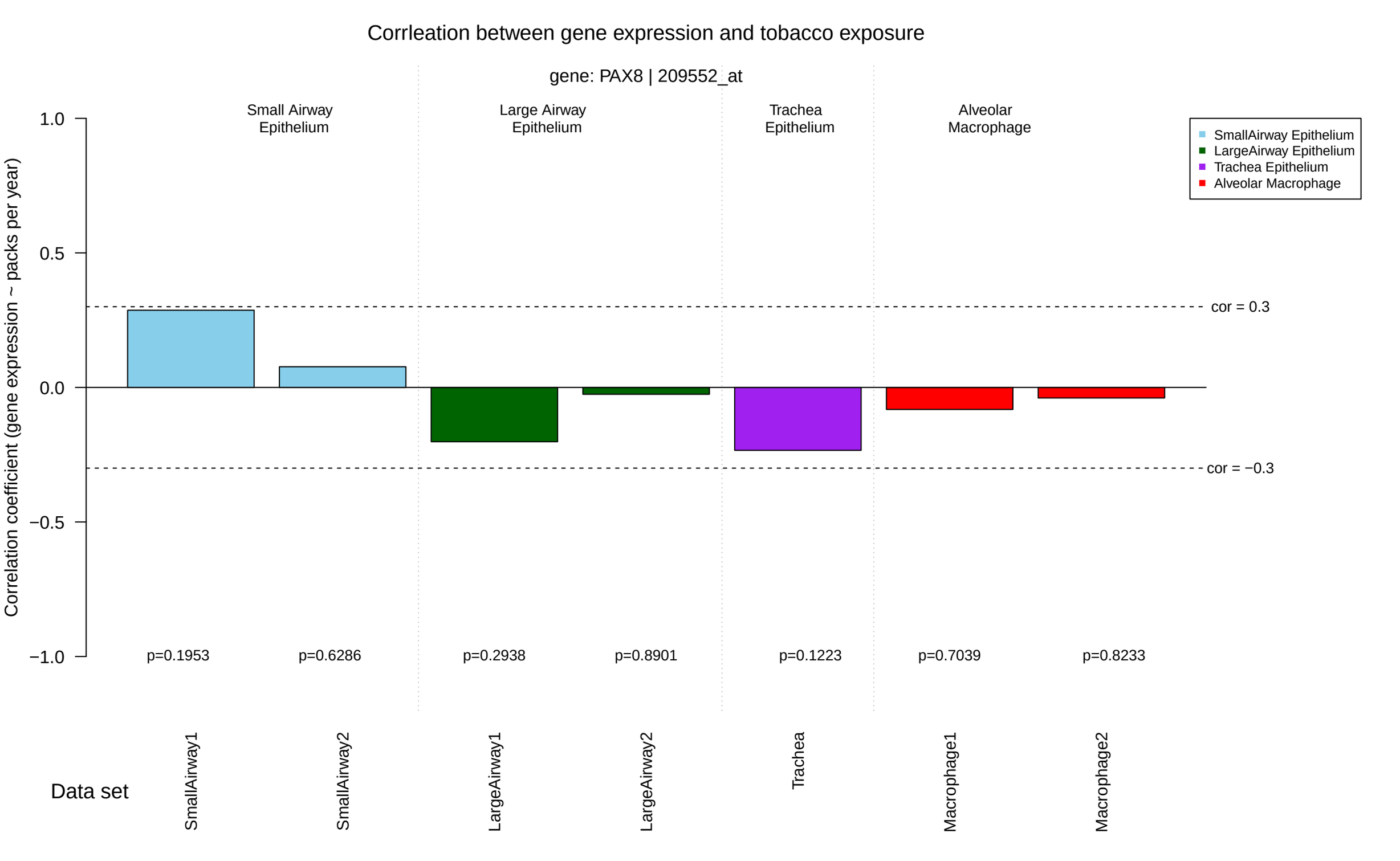

set: LargeAirway1 (p=0.9297)
PAX8 | 209552_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.593)
PAX8 | 209552_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.167)
PAX8 | 209552_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6151)
PAX8 | 209552_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5218)
PAX8 | 209552_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAX8 | 209552_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1953
p=0.6286
p=0.2938
p=0.8901
p=0.1223
p=0.7039
p=0.8233
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2